Decoding the Curb

Data-Driven Analysis of NYC Parking Violations

Data Visualization & Analytics Project

Newton School of Technology

Team ID: G-12

Team Members: Kavya Katal, Abuzar Haideri, Ayush Tiwari, Vaibhav Rajput, Abhay Malik, Deepak Pandey

Faculty Mentor: Mr. Archit Raj

□ The Problem

The Problem

NYC issues millions of parking tickets annually

Enforcement follows fixed patterns, not real demand

Leads to:

- Over-enforcement of low-risk violations
- Under-enforcement of safety-critical zones

Core Question

👉 **Is NYC parking enforcement optimized or inefficient?**

Decision Supported:

Content publishing & optimization strategy

💡 Enforcement is reactive, not data-driven. Fixed schedules miss real violation hotspots.

□ Objective

Identify inefficiencies in enforcement system

Analyze violation trends across:

Time

Seasonal, monthly, and daily patterns

Boroughs

Geographic distribution and hotspots

Vehicle Types

Make, body type, and age patterns

Segment Repeat Offenders

Identify high-frequency violator profiles and patterns

Data-Driven Recommendations

Provide actionable insights for enforcement optimization

□ Approach / Methodology

~12,000	29	9	3
Records Analyzed	Raw Columns	Engineered Features	Tools Used
NYC Open Data (FY 2014)	Original dataset fields	Custom KPI columns created	Python, Tableau, Statistics

Python ETL Pipeline

01

Data Cleaning

Null handling, deduplication, type casting

02

Feature Engineering

9 new KPI columns created

03

Statistical Analysis

ANOVA, Chi-Square, Regression

Tools Used

Python (Pandas)

Tableau

Statistical Analysis

□ Data Overview

29 → 9

Raw columns → Engineered features

Dataset transformed through cleaning and feature engineering to extract actionable insights from NYC parking violation records.

Key Fields

- **Plate ID**
- **Violation Code**
- **Borough**
- **Fine Amount**

Created Features

Offender Tier
Frequency-based violator classification

Violation Severity
Safety vs administrative ranking

Vehicle Age
Derived from vehicle year

Avoidable Violations
Preventable with better signage

📊 Feature engineering transformed raw violation data into actionable business intelligence metrics.

□ KPI Framework

Helps measure efficiency, safety, and revenue

Total Tickets Issued

Volume metric for enforcement activity measurement

Total Revenue

Financial impact of parking enforcement operations

Average Fine

Mean penalty amount across all violations

Repeat Offender Rate

Percentage of vehicles with multiple violations

Safety Violation %

Share of high-risk safety-related violations

Avoidable Violation %

Preventable violations with better signage

📈 **Framework captures efficiency, safety, and revenue dimensions to prevent optimizing wrong metrics.**

□ EDA: Trends & Time

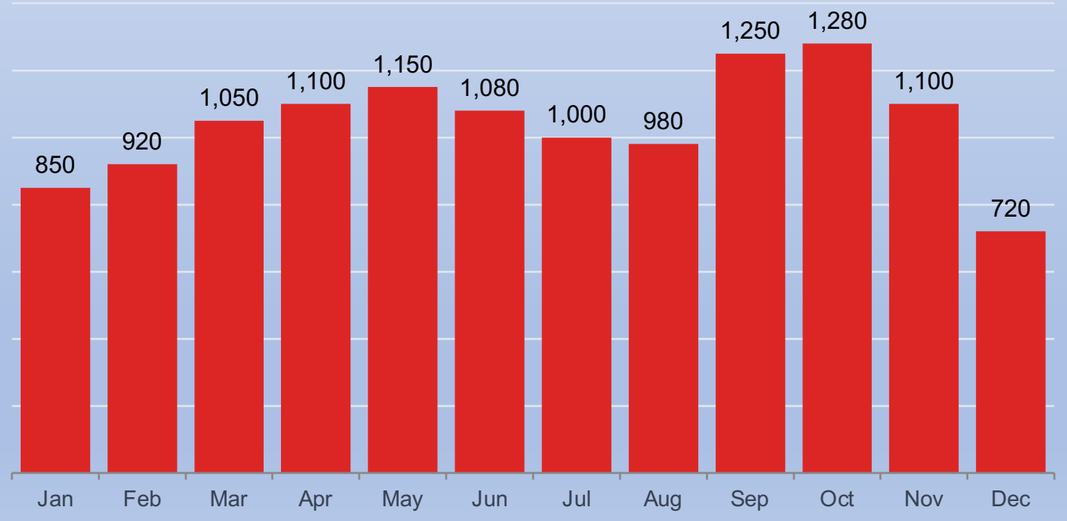
Monthly Patterns

Peak months: September – October

Lowest: December

Seasonal spikes correlate with:

- Back-to-school traffic
- Holiday shopping rush
- Street cleaning schedules



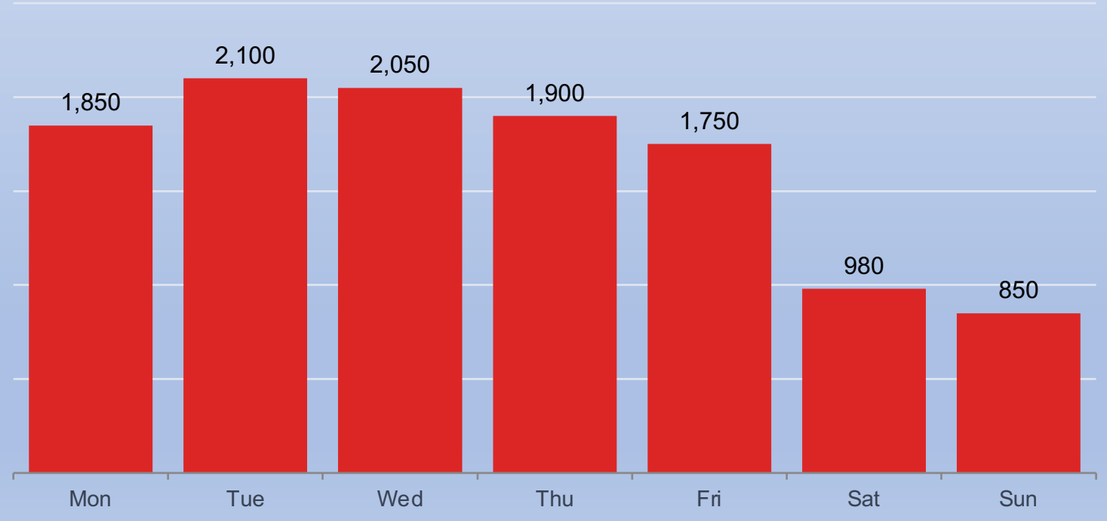
Day-of-Week Patterns

Highest: Tuesday & Wednesday

Lowest: Weekend

Weekday concentration indicates:

- Street cleaning enforcement
- Commuter parking pressure
- Reduced weekend patrols



👉 **Enforcement linked to street cleaning schedules — not real-time demand patterns.**

□ EDA: Borough Insights

Manhattan

High-value tickets

Premium parking zones with highest average fine amounts per violation

Queens & Brooklyn

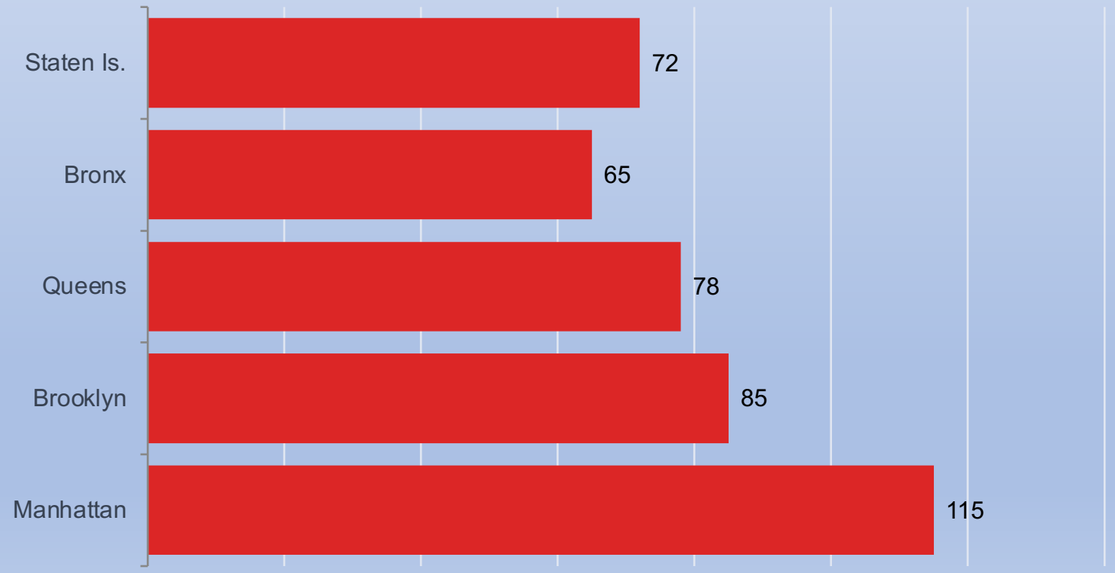
High volume tickets

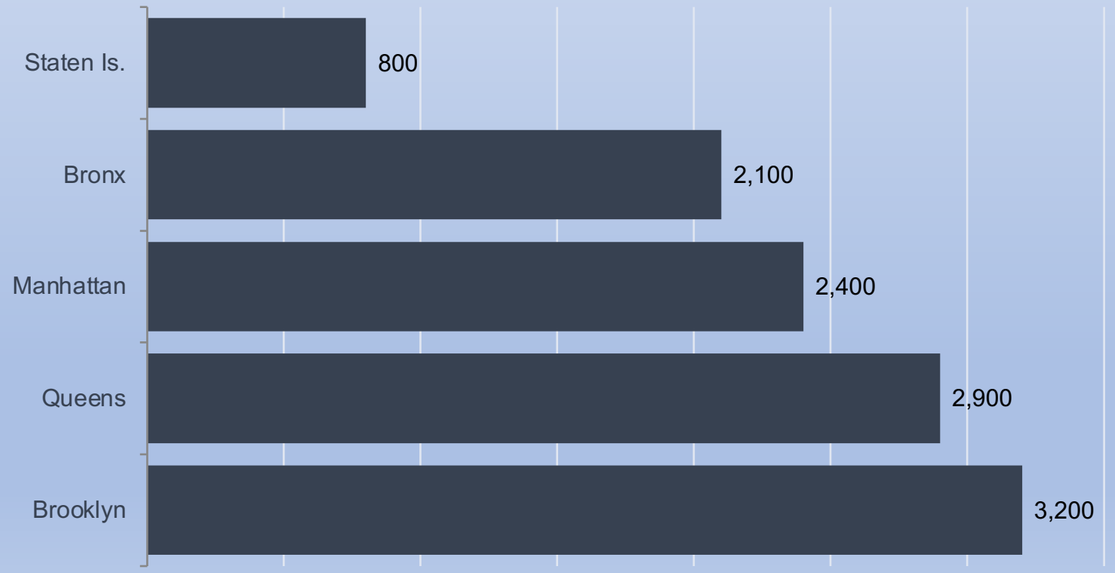
Residential and commercial areas with highest ticket issuance counts

Bronx

Administrative violations

Higher proportion of paperwork and registration-related violations





👉 **Each borough has a unique violation pattern — one-size-fits-all enforcement is suboptimal.**

□ Key Insight: Pareto Principle

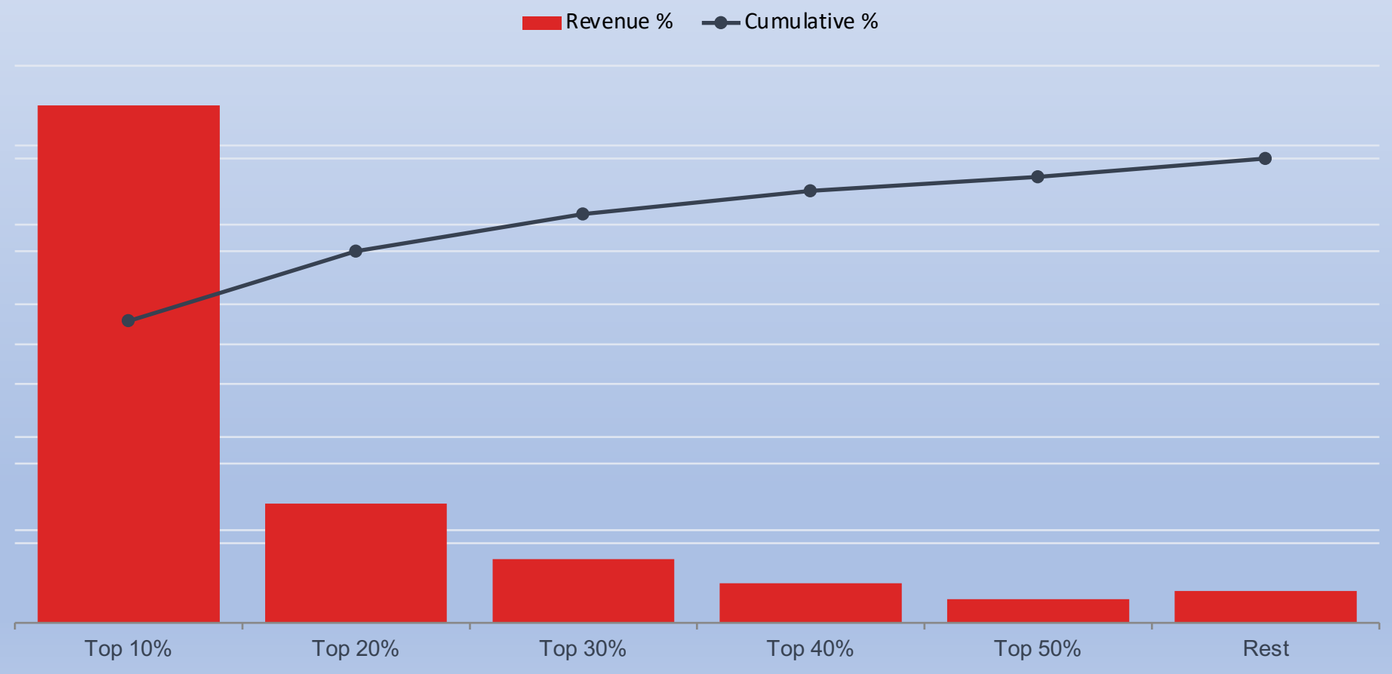
20%

of vehicles

→

~80%

of revenue



Repeat offenders dominate ticket volume

A small subset of vehicles generates the majority of enforcement revenue

What This Means

❌ System acts like a **subscription model**, not deterrence

❌ Repeat offenders are **not penalized progressively**

❌ Revenue focus **overrides safety goals**

✅ Opportunity for **targeted intervention**

□ Statistical Insights

Statistical tests reveal significant patterns in enforcement data

Chi-Square Test

Significant

Borough differences in violation types are statistically significant — enforcement is not uniform across NYC.

ANOVA Test

Significant

Fine amounts vary significantly across boroughs — Manhattan fines are materially higher than other areas.

Regression

Weak predictive power

Linear models show weak predictive relationship — current factors don't explain violation patterns well.

📊 **Key Takeaway:** The system is not adaptive — it follows fixed rules rather than responding to data-driven signals.

Test	Result	Insight	Implication
Chi-Square	Significant	Borough differences exist	Need borough-specific strategies
ANOVA	Significant	Fine amounts vary by area	Pricing is not standardized

□ Recommendations

Data-driven strategies for smarter enforcement

01 Repeat Offender Escalation

Increase fines for frequent violators

Implement tiered penalty system based on violation history

Expected: +20% deterrence effect

02 Smart Digital Signage

Reduce avoidable violations

Deploy dynamic parking signs with real-time rule updates and alerts

Expected: -15% avoidable tickets

03 Shift Optimization

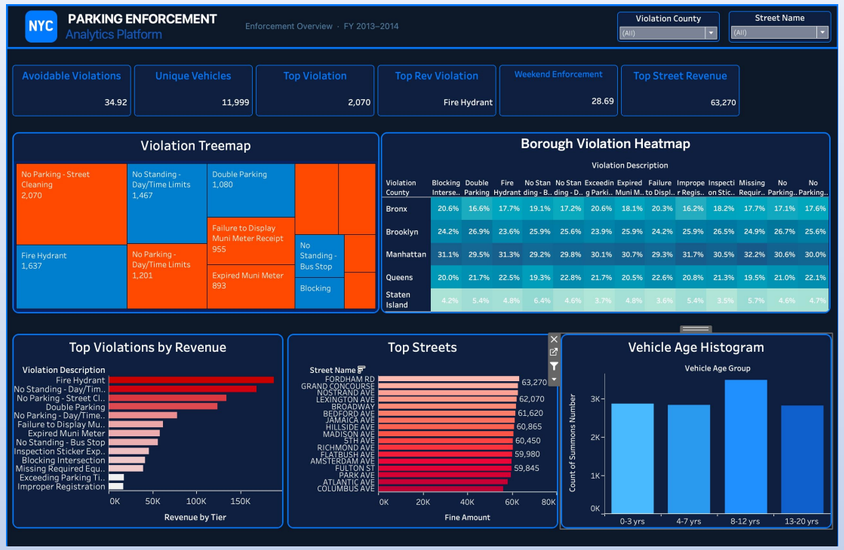
Increase weekend enforcement

Reallocate patrol resources from low-activity weekdays to weekends

Expected: Better coverage

💡 **All recommendations are backed by EDA findings and statistical validation from the dataset.**

□ Tableau Dashboard



□ Conclusion

NYC system is efficient but not smart — focuses on revenue, not safety.

Improve Safety

Shift focus to safety-critical violations rather than revenue-generating minor offenses

Increase Efficiency

Data-driven patrol scheduling aligned with actual violation patterns and hotspots

Reduce Unnecessary Violations

Smart signage and digital alerts to prevent avoidable parking mistakes

Future Scope

🔮 **Predictive analytics** — ML models to forecast violation hotspots before they occur

🔮 **Smart enforcement** — IoT sensors and automated violation detection systems

🔮 **Real-time dashboards** — Live monitoring for dynamic resource allocation

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